

Usability vs. Security: A Measurement Study on Multi-Factor Authentication Adoption and Its Real-World Impact

Kelly Chung
MICS, School of Information
University of California Berkeley
kellychung@berkeley.edu

Gregory Mamoyac
MICS, School of Information
University of California Berkeley
gregmamoyac@berkeley.edu
Adedayo Odugbesan
MICS, School of Information
University of California Berkeley
abubakry@berkeley.edu

Syed Muhammad
MICS, School of Information
University of California Berkeley
smuhammad@berkeley.edu

Abstract—This study will analyze self-reported user experiences with MFA through survey data and statistical analysis to determine how usability influences adoption rates. The findings will guide recommendations for making MFA more user-friendly without significantly reducing security.

Keywords—Multi-Factor Authentication, Usability, Security, Trust, MFA Solutions, Adoption,

I. INTRODUCTION

In an increasingly digitized world, the importance of robust security measures is paramount. Multi-Factor Authentication (MFA) has emerged as a critical tool for safeguarding online accounts. This study investigates the usability factors influencing the adoption of MFA, focusing on how convenience, perceived complexity, and trust in technology affect user experiences. Despite the recognized importance of MFA in enhancing security, many users still refrain from adopting these measures, often citing usability concerns. By analyzing self-reported user experiences with different MFA methods, this research aims to identify key areas for improvement in MFA design that balance usability and security. The findings are expected to provide actionable recommendations for refining MFA systems, making them more user-friendly, and ultimately increasing their adoption rates across various demographics.

II. EXPERIMENTAL SPECIFICATIONS

To explore the usability of MFA and its impact on user adoption, a mixed-methods study conducted involving a total of 90 participants divided into pilot and test groups. The study utilizes an online survey distributed through Prolific and Qualtrics platforms. The survey measures factors such as the frequency of MFA use, preferences for different MFA methods and perceptions around usability and security. Quantitative data will be collected using scale ratings, while qualitative data will be gathered through open-ended responses. This approach aims to capture user sentiments, frustrations, and preferences to comprehensively understand the trade-offs users make between usability and security. The results will inform the development of more effective authentication methods, leading to increased user adoption and improved security practices in real-world applications.

A. Prolific Recruitment

Participants were recruited to ensure a diverse demographic, including variations in technology comfort

levels and backgrounds who were most familiar with MFA in personal and professional settings. This was done to sample a more accurate group of technology users that have used MFA and can provide the most useful research information.

The Prolific research platform allows for participant selection through the Screening module available in study setup.

III. PREPARE YOUR PAPER BEFORE STYLING

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections A-D below for more information on proofreading, spelling and grammar.

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Number equations consecutively. Equation numbers, within parentheses, are to position flush right, as in (1), using a right tab stop. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \quad (1)$$

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- The word “data” is plural, not singular.
- The subscript for the permeability of vacuum μ_0 , and other common scientific constants, is zero with subscript formatting, not a lowercase letter “o”.
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
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- In your paper title, if the words “that uses” can accurately replace the word “using”, capitalize the “u”; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones “affect” and “effect”, “complement” and “compliment”, “discreet” and “discrete”, “principal” and “principle”.
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Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced. Styles named “Heading 1”, “Heading 2”, “Heading 3”, and “Heading 4” are prescribed.

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TABLE I. TABLE TYPE STYLES

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
copy	More table copy ^a		

^a Sample of a Table footnote. (Table footnote)

Fig. 1. Example of a figure caption. (figure caption)

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when

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ACKNOWLEDGMENT (*Heading 5*)

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REFERENCES

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Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

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- [1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955. (*references*)
- [2] J. Clerk Maxwell, *A Treatise on Electricity and Magnetism*, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
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- [4] K. Elissa, “Title of paper if known,” unpublished.
- [5] R. Nicole, “Title of paper with only first word capitalized,” *J. Name Stand. Abbrev.*, in press.
- [6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, “Electron spectroscopy studies on magneto-optical media and plastic substrate interface,” *IEEE Transl. J. Magn. Japan*, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetism Japan, p. 301, 1982].
- [7] M. Young, *The Technical Writer’s Handbook*. Mill Valley, CA: University Science, 1989.

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